

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Radian Measure

Radian questions are sector questions: angle, radius, arc, area.

RADIAN MEASURE · P1 · 10

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA11 website answer bank. The topic reduces to these exam moves.

01 RAD: Degrees To Radians

TRIGGER *convert degrees/radians*

02 ARC: Arc Length

TRIGGER *arc length or perimeter of sector*

03 SECT: Sector Area

TRIGGER *sector area*

04 SEG: Segment Problems

TRIGGER *minor segment, shaded area*

BANK EVIDENCE Local pages: 26 problem, 38 answer, 14 notes. Website primary entries: 18.

01 RAD: Degrees To Radians

TRIGGER *convert degrees/radians*

BECOMES Use π radians equals 180° .

FIRST LINE TO WRITE

$$180^\circ = \pi \text{ radians}$$

SIMPLEST STRATEGY

- 1 Read the starting unit.
- 2 Apply the conversion factor.
- 3 Deliver exact multiples of π where possible.

WORKED MODEL

$$60^\circ = 60 \cdot \frac{\pi}{180} = \frac{\pi}{3}.$$

HARD VARIANTS

- 1 Convert to radians before using sector formulas.
- 2 Keep exact multiples of π until the final answer.
- 3 If an angle is reflex or major, decide whether to use θ or $2\pi - \theta$.

— BOTTOM LINE

Keep exact π unless decimals are requested.

02 ARC: Arc Length

TRIGGER *arc length or perimeter of sector*

BECOMES Use $s = r\theta$ with θ in radians.

FIRST LINE TO WRITE

$$s = r\theta$$

SIMPLEST STRATEGY

- 1 Angle must be in radians.
- 2 Radius identified.
- 3 Calculate arc, then add straight sides if perimeter.

WORKED MODEL

$$r = 5, \theta = 1.2 \Rightarrow s = 6.$$

HARD VARIANTS

- 1 For perimeter, add the two radii after finding the arc length.
- 2 If given chord/sector information, find θ first in radians.
- 3 Use major/minor wording to choose the correct angle.

— BOTTOM LINE

Arc length uses radians only.

03 SECT: Sector Area

TRIGGER *sector area*

BECOMES Use $\frac{1}{2}r^2\theta$.

FIRST LINE TO WRITE

$$A = \frac{1}{2}r^2\theta$$

SIMPLEST STRATEGY

- 1 Secure radians.
- 2 Enter radius squared.
- 3 Calculate sector area.
- 4 Take away triangle if segment is needed.

WORKED MODEL

$$A = \frac{1}{2}(4)^2 \left(\frac{\pi}{3}\right) = \frac{8\pi}{3}.$$

HARD VARIANTS

- 1 Sector area uses $\frac{1}{2}r^2\theta$, never degrees.
- 2 For shaded regions, compute sector and triangle separately before subtracting.
- 3 If radius or angle is unknown, set up the sector formula and solve symbolically.

— BOTTOM LINE

Sector area is half $r^2\theta$, not $\pi r^2\theta$.

Practice 03

SAME IDEA Use SECT: Sector Area.

QUESTION

| Find sector area for $r = 6, \theta = 2$.

YOUR SOLUTION

04 SEG: Segment Problems

TRIGGER *minor segment, shaded area*

BECOMES Sector minus triangle.

FIRST LINE TO WRITE

$$\text{segment} = \frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin \theta$$

SIMPLEST STRATEGY

- 1 Sector area first.
- 2 Evaluate triangle area.
- 3 Get difference.

WORKED MODEL

$$\text{segment} = \frac{1}{2}r^2(\theta - \sin \theta).$$

HARD VARIANTS

- 1 Minor segment is sector minus triangle; major segment is circle minus minor segment.
- 2 Use $\frac{1}{2}r^2 \sin \theta$ for the triangle only when θ is included.
- 3 Check whether the shaded part is the small segment or its complement.

— BOTTOM LINE

Most shaded segment questions are subtraction questions.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
convert degrees/radians	$180^\circ = \pi \text{ radians}$
arc length or perimeter of sector	$s = r\theta$
sector area	$A = \frac{1}{2}r^2\theta$
minor segment, shaded area	$\text{segment} = \frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin \theta$